

UNIAXIAL COMPRESSIONAL STRESS-STRAIN RELATION OF CONCRETE

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INTRODUCTION

The stress-strain relation of concrete under compression is known to contain an ascending portion and a descending portion. The ascending portion is well characterized. However, the descending portion has not yet been completely developed. Since the relations for both portions are steeper as the ultimate concrete strength becomes higher, concrete with higher strength has higher potential of catastrophic failure after the peak strength is reached. Therefore, the descending portion is important if ductility of the concrete members is desired. Development of an accurate stress-strain relation for the descending portion becomes necessary. Many equations have been suggested to represent the relation for this portion. An excellent survey for the entire range of developed stress-strain relation for concrete was given by Aoyama and Noguchi (1979). Among the developed relations, some are simpler in forms or more accurate in results than the others. Nevertheless, when it comes to large strain, say 1% or greater, the suggested equations generally do not present favorable comparisons with test data. Lack of sufficient test data in the descending portion is possibly the reason.

This note proposes a new equation to represent the uniaxial compressional stress-strain relation of concrete. This equation may be considered as a generalized form of Popovics (1970) and Saenz (1964). It consists of two parameters, one to control the ascending behavior and the other to control the descending behavior. These two parameters can be used to refine the accuracy of the stress-strain relation when more test data are available. At this time, the result of Wang et al. (1978) will be applied to illustrate the accuracy of the proposed equation.

SUGGESTED STRESS-STRAIN RELATION

The recommended stress-strain relation of concrete is

$$y = \frac{mx}{1 + \left(m - \frac{n}{n-1}\right)x + \frac{x^n}{n-1}} \dots\dots\dots (1)$$

where $y = f_c/f'_c$ = the ratio of the concrete stress to the ultimate strength; $x = \epsilon/\epsilon_c$ = the ratio of the concrete strain to the strain at $y = 1$; $m = E_o/E_c$ = the ratio of the initial tangent modulus to the secant modulus at $y = 1$; and n = a factor to control the steepness rate for the descending portions

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of the stress-strain relation. The variation of m is equivalent to control the steepness rate of the ascending portion of the relation since it adjusts the modulus ratio in the ascending zone. For any given parameters m and n , Eq. 1 always satisfies the basic conditions of a complete stress-strain relation that it ascends when $\epsilon < \epsilon_c$ and descends when $\epsilon > \epsilon_c$. When $\epsilon = \epsilon_c$, the stress $fc = f'_c$, and it is the peak value in the entire range of the relation. When the strain goes to a very large value, the stress goes to zero as anticipated, provided that $m(n - 1) > n$ and $n > 1$.

Eq. 1 becomes the relation of Popovics (1970) if the parameter m is set equal to $n/(n - 1)$. It also becomes the form of Saenz (1964) if $n = 2$, although the relation of that reference is only applicable to the ascending portion. Therefore, Eq. 1 may be considered to be the generalized relation of Popovics (1970) and Saenz (1964).

DETERMINATION OF PARAMETERS m AND n

To illustrate the accuracy of Eq. 1, data given by Wang et al. (1978) are applied. Based on test data, that reference expresses the stress-strain relation by using the ratio of two quadratic forms

$$y = \frac{Ax + Bx^2}{1 + Cx + Dx^2} \dots \dots \dots (2)$$

where A , B , C , and D = the constants determined from test data. Eq. 2 is applicable to both the ascending and descending portions, but with different sets of constants. By combining the parameter suggested by Popovics (1970) and the result calculated by using Eq. 2, the parameters m and n of Eq. 1 are given by

$$m = 1 + \frac{17.9}{f'_c} \quad (f'_c \text{ in MPa}) \dots \dots \dots (3a)$$

$$m = 1 + \frac{2,600}{f'_c} \quad (f'_c \text{ in psi}) \dots \dots \dots (3b)$$

$$n = \frac{f'_c}{6.68} - 1.85 > 1 \quad (f'_c \text{ in MPa}) \dots \dots \dots (4a)$$

$$n = \frac{f'_c}{970} - 1.85 > 1 \quad (f'_c \text{ in psi}) \dots \dots \dots (4b)$$

Upon substituting Eqs. 3a, 3b, 4a, and 4b into Eq. 1, an explicit form of stress-strain relation of concrete is obtained once the ultimate strength, f'_c , is given.

It is noted by using the initial modulus for normal-weight concrete, the secant modulus, and the strain at $fc = f'_c$ can be obtained by $E_c = E_o/m$ and $\epsilon_c = f'_c/E_c$. Incidentally, these results are very close to the results of Popovics (1970).

COMPARISONS

The results obtained from Eq. 1 are compared to the results of Wang et al. (1978), which includes examples of $f'_c = 20.7$ Mpa (3 ksi) to 89.6 Mpa

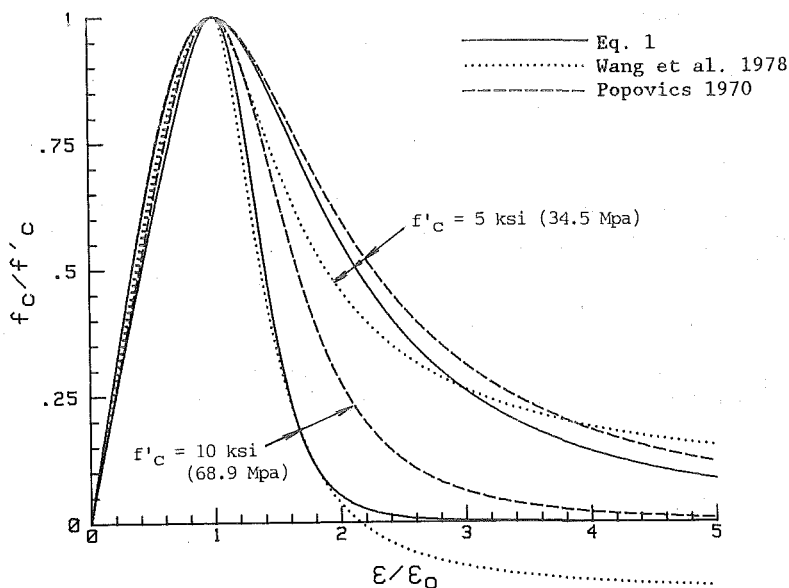


FIG. 1. Compressive Stress-Strain Relation of Concrete

(13 ksi) at an interval of 6.9 Mpa (1 ksi). For the ascending portion, excellent agreements in using different formulations are obtained for all cases. For the descending portion, the comparisons are generally very favorable to Eq. 1 with the parameters of Eqs. 3a, 3b, 4a, and 4b, although some discrepancies in the results are observed for strength lower than 34.5 Mpa (5 ksi). The relations for $f'_c = 34.5$ Mpa (5 ksi) and $f'_c = 68.9$ Mpa (10 ksi) are depicted in Fig. 1, showing excellent correlations between Eq. 1 and Wang et al. (1978), especially for the range of strength higher than 34.5 Mpa (5 ksi). At the range of large strain, the present relation provides a near-zero strength while Wang et al. (1978) provides negative strength for some cases, which cannot be true in the reality.

Also shown in Fig. 1 are the results by using the formula of Popovics (1970). Although, all the relations come out to be in the same trend, the results of that reference are not as close as that of Eq. 1 to the results of Wang et al. (1978).

CONCLUSIONS

1. The suggested stress-strain relation of concrete correlates very well with the test data, including both ascending and descending portions. It is particularly favorable when the strength is higher than 34.5 Mpa (5 ksi).

2. The suggested relation is applicable throughout the entire range of strain without changing the coefficients.

3. The suggested relation provides two parameters, one to adjust the ascending portion and the other to control the descending portion. These parameters can be modified when more test data are available.

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APPENDIX. REFERENCES

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